Probehead operating at 35 GHz for continuous wave and pulse electron paramagnetic resonance applications

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The design, construction, numerical modeling, and performance of a probehead for electron paramagnetic resonance (EPR) spectroscopy at 34–36 GHz is described. A classical cylindrical cavity operating in the TE_{011} mode with adjustable frequency and coupling has been found to be well suited for continuous wave and pulse EPR studies of frozen solutions of transition metal complexes at low temperature. The highest attention is given to the probehead performance in the pulse mode. The implemented design has been analyzed in detail using numerical modeling. The distribution of the electromagnetic fields, eigenfrequencies, quality factors, coupling coefficients, and conversion factors are calculated and compared with experimental data. The results demonstrate the effectiveness of the design and can serve as a guide for probehead optimization. © *2006 American Institute of Physics.* [DOI: 10.1063/1.2216849]

INTRODUCTION

Pulse electron paramagnetic resonance (EPR) spectrometers operating at frequencies around 35 GHz have become popular in recent years, because of the large g value and orientation selectivity, the reduced second-order effects, and the suitability for echo modulation measurements of many types of nuclei.[1–4] The design and construction of probeheads operating in this frequency range has therefore turned out to be a key issue for EPR spectroscopists. Versatility, performance, sensitivity, and a convenient handling are major criteria that have to be optimized.

During the last decades a large variety of resonators have been developed for EPR spectroscopy.[5] We only mention some representative designs. "Lumped-mode" structures, such as loop-gap resonators[6–8] (LGR) and bridged loop-gap[9] resonators are widely used at L, S, and X bands (bands are given in accordance with IEEE standard[10]). Their key advantages are a large bandwidth and a high conversion factor $\Lambda = B_1/\sqrt{PQ}$, where B_1 is the magnetic induction in the frame rotating at a given microwave (mw) frequency, P is the incident (or emitted) mw power, and Q is the quality factor of the resonator. Strongly overcoupled high-Q dielectric ring (DR) resonators operating in the $TE_{01\delta}$ mode at X band[11] are frequently used in pulse EPR experiments. They are distinguished by robustness, frequency stability, and high Λ values. Cylindrical TE_{011} cavities (CC) are exploited at Ka band[12] and are also favorite structures at W-band frequencies.[13] Simple design, wide frequency tuning, and intrinsically high Q values are attractive features for this type of resonator. A few LGR (Refs. 14 and 15) and DR (Ref. 4) applications at Ka band are reported as well.

Cylindrical (or rectangular) cavities have, in general, smaller Λ values than LGRs and DRs. However, this disadvantage becomes less pronounced when sample space has to be enlarged to achieve a measurable signal from samples with a low concentration of paramagnetic centers. A typical example is a frozen solution of a protein containing transition metals. In such a case the molar sensitivity is a key factor which determines applicability of the advanced pulse EPR techniques used to obtain high-resolution spectra. Reported investigations show that even at X band a LGR with 5 mm sample loop and Varian TE_{102} cavity are comparable in efficiency.[6] On the other hand, a power level provided by modern solid state power amplifiers is sufficient to drive low Λ resonators effectively in the case of pulse EPR. Rectangular cavities have typically higher conversion factors (normalized on Q) than cylindrical resonators,[5] however, they are more sensitive to the load and less convenient for multipurpose applications. This make CCs popular at Ka-band frequencies for cw EPR, pulse electron nuclear double resonance (ENDOR), and more advanced pulse EPR techniques.[2,12,16–19] We have successfully used a home-built probehead based on CC at Ka band for a variety of pulse experiments including two-dimensional time-domain correlation techniques with matched pulses[20] or hyperfine decoupling.[21]

There are several approaches to couple a cylindrical TE_{011} cavity to a rectangular waveguide. A CC coupled through the iris in the planar topwall (off center) is used in the Varian wire-wound cavity and the "ribbon-wound" resonator.[16] A cylindrical sidewall to waveguide narrow wall junction is implemented by Sienkiewicz *et al.*,[12] and a bottom-to-widewall configuration is exploited in the ER5106 probehead from Bruker BioSpin. For ENDOR experiments a hair-pin coil inside the cavity has been used in these designs. We chose the Varian approach which allows one to position the coupling window in the vicinity of the maximum field in the resonator. This design does not require a waveguide short, and offers sufficient space around the resonator cylin-

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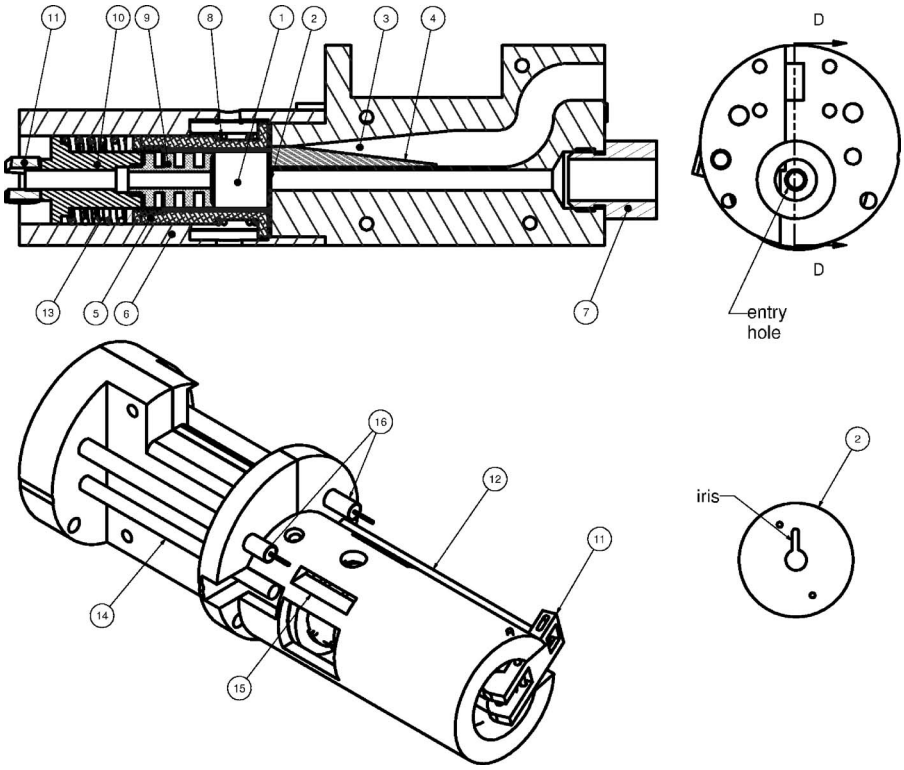


FIG. 1. Cavity assembly and coupling structure. (1) Cylindrical cavity with 11 mm inner diameter; (2) diaphragm with rectangular iris; (3) coupling structure; (4) teflon or PEEK movable wedge; (5) cavity housing; (6) housing; (7) sample holder guide; (8) rf saddle coil; [(9) and (10)] movable short; (11) lever; (12) frequency tuning rod; (13) spring which provides return movement of the short; (14) coupling tuning rod; (15) slits where modulation coils are fixed; (16) rf coaxes. The components are shown in scale.

der to mount rf and modulation coils. Since we are mainly interested in the mw pulse performance, the ENDOR coil was placed outside the cavity. The described probehead was designed for a pulse spectrometer operating at 35 GHz.[3] We believe that the presented detailed characterization of a particular CC implementation can help to optimize newly designed probeheads.

PROBEHEAD DESIGN

The design of the probehead, shown in Fig. 1, is based on a cylindrical cavity with an inner diameter of 11 mm (1), coupled on its top via a diaphragm (2) to the waveguide, and a Gordon coupling structure (3) with a movable teflon or high-strength polymer (PEEK, Angst+Pfister) wedge (4). The coupling structure consists of a specially shaped waveguide, with the height of the WR28 waveguide, extruded into a solid block. It serves as the basis to assemble the housings [(5) and (6)], the rf coaxes (15), the rf matching circuits, and the tuning rods. It also provides a guide (7) for the sample holder to allow for a precise positioning of sample tubes with outer diameters up to 4 mm. The body of the cavity (1) can easily be exchanged.

For pulse experiments a full body copper cylinder can be used, whereas for cw EPR the cavity wall has to be transparent for the 100 kHz modulation field, and for ENDOR experiments for the rf up to about 100 MHz. In the latter case, the cavity, which is made of a quartz tube (Wilmad), is silver and gold plated on its inner surface. The gold layer prevents the silver film from damage caused by temperature changes between liquid helium and room temperature. This layer was made by firing gold paint (Heraeus, GG Q 3%–15%) onto the surface, following the procedure described by Pfenninger *et al.*[9] Then the silver layer was added by electroplating. The total thickness of the metallic film was in the range of $2-5\ \mu$m. In the middle of the resonating volume the metallic layer is interrupted by eight circular slits (width 0.1 mm, separation 0.4 mm) to make the resonator transparent to rf fields. A four-turn saddle coil (8) is wrapped around the cavity housing (5).

A movable short [9 and 10] allows for a frequency tuning of about 2 GHz for the TE_{011} mode. It consists of two parts, a silver-plated alumina cylinder (9) with three 2-mm-wide grooves separated by 3.8 mm. The lower part of the short (10) fits to the bottom of the housing (6) and guides the short along the common axis of the cavity, housing (5), and assembly body (6). A lever (11), which is fixed on the housing (6), translates a movement of the frequency tuning rod (12) with a reduction coefficient of 1.5 in opposite displacement of the short. The spring (13) allows for the return movement of the short, has a guiding effect, and damps unwanted vibrations. The rod (14) is used to move the plunger (4). The exchangeable modulation coils (not shown) are fixed in slits (15). The parts (5) and (6) and (10) and (11) are made of high-strength polymer (PEEK, Angst+Pfister). The remaining components are fabricated of aluminum and silver plated.

The frequency tuning assembly allows for a variation of the length of the cavity by about ±3 mm, typically from 6 to 12 mm. This is sufficient to correct for the frequency shift in a wide range of situations, such as different sizes and types of sample tubes, different dielectric properties of the samples, etc. Nevertheless, we made use of two configurations, one with a cavity length of 9 ± 3 mm and one with 12 ± 3 mm using shorts [(9) and (10)] of different length. The first configuration is used for 2.9–3.8 mm quartz tubes

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TABLE I. Mode selection.

	Tube o.d. (mm)	Tube i.d. (mm)	Cavity length (mm)	Mode
1	2.9–3.8	2–3.4	6–12	TE_{011}
2	3.8	2.9–3.4	9–15	TE_{012}
3	None	None	9–15	TE_{011}

(TE_{011}) and the second for situations without sample tube (TE_{011}), and for 3.8 mm tubes (TE_{012} mode), as shown in Table I.

The cavity assembly is connected to a *H*-plane 90° waveguide turn soldered to the top of the probehead via a WR28 thin-walled stainless steel waveguide plated with silver on the inner surface. The probehead top fits to the CF935 cryostat from Oxford and does not have any special features to be described.

MODELING

The influence of the coupling structure, entry holes, and size and dielectric properties of the sample tube on the cavity eigenmodes and the electromagnetic field distribution were numerically simulated using the "Microwave Studio" package (CST). The "Time Domain Solver" module has been used in all the simulations, and a cavity diameter of 11 mm is assumed for all presented data. The primary goal of these simulations was to optimize the geometry of the cavity and coupling structure for operation in the TE_{011} mode and to provide stable, easily reproducible operational conditions for a wide range of dielectric loads with the upper limit restricted to standard 3.8 mm quartz tubes usually employed at *X*-band frequencies. Simulations and measurements were performed for two setups, which differ in the sample access hole (∅3.1 and ∅4 mm), referred further on as 3 and 4 mm sets. The first set serves for test measurements and the second is continuously in use. The measurements were performed with the vector network analyzer HP 8722ES.

The eigenfrequencies of the resonator were calculated as a function of the length and the dielectric load, using the model shown in Fig. 2. The empty cavity without sample access hole exhibits several resonance modes, when the length was varied from 6 to 12 mm [Fig. 3(a)]. The cavity was excited through the small iris with a Gaussian-shaped pulse covering the frequency range from 30 to 40 GHz. Because of the symmetry of the problem, the *xz* plane was included as an additional boundary condition. Solid lines in the figure represent the known analytical solutions for a frequency f of the eigenmode TE/TM_{mnp} in the empty cylindrical cavity,[5]

$$\left(\frac{f}{c}\right)^2 = \left(\frac{1}{\lambda_{mn}}\right)^2 + \left(\frac{p}{2l}\right)^2, \tag{1}$$

where λ_{mn} is the mode-dependent cutoff wavelength, l is the length of the cavity, and the mode indices m, n, and p denote the numbers of half-wave variations along the azimuthal angle ϕ, the radius r, and the length l, respectively.

As a numerical example, at 35 GHz the empty cavity (r=5.5 mm) has to have a length of 13.6 mm to resonate in the TE_{011} mode. Increasing the radius r of the cavity leads to a linear increase in λ_{mn}, with factors 3.42, 2.06, and 1.64 for the TE_{11p}, TE_{21p}, and TE_{01p} modes, respectively,[5] and consequently to a decrease of the resonance length for a given frequency. When the radius increases, all the modes are shifted to lower frequencies, but the TE_{112} mode experiences only a small shift as expected from Eq. (1). At r=5.7 mm, a TE_{011}/TE_{112} mode crossing is observed for 35.9 GHz and at r=5.5 mm the crossing occurs at 37.3 GHz [Fig. 3(a)].

The TE_{011}/TE_{111} mode degeneracy is lifted by the sample access hole. The dielectric load influences the resonance frequencies in an opposite way to their dependence on the cavity radius. The simulated eigenfrequencies of the resonator loaded with the quartz tube [outer diameter (o.d.) 2.9 mm, inner diameter (i.d.) 2 mm] are in agreement with the measured frequencies shown in Fig. 3(b). In comparison

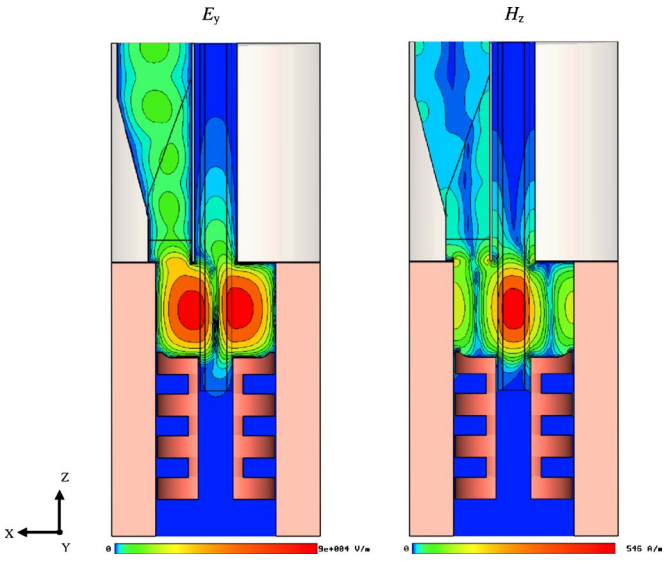


FIG. 2. (Color online) Simulated electromagnetic field distribution in the cavity and coupling structure on resonance and at critical coupling. The magnitude values for the $\mathbf{E}_y$ and $\mathbf{H}_z$ components are shown. The wave is propagating in the coupling structure and is standing in the resonator. The entry hole has a diameter of 4 mm, the sample tube o.d. is 2.9 mm, and the teflon plunger is placed 2 mm away from the coupling iris with a height of 1.2 mm. A logarithmic scale is used. A microwave power of 1 W was used providing a maximal magnetic field of 546 A/m.

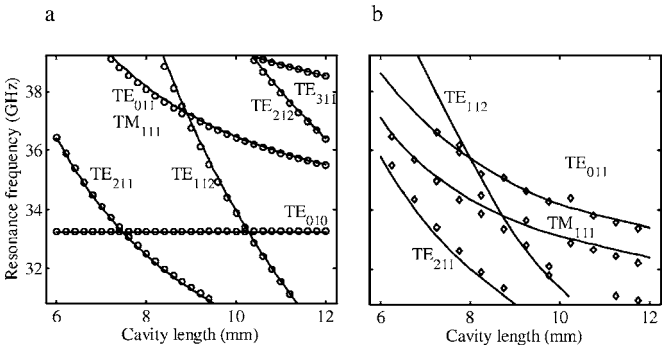


FIG. 3. Numerical simulation of the eigenmodes. (a) Empty cavity. (b) Cavity loaded with a 2.9 mm o.d. quartz tube. In (a) circles represent the numerical solutions and solid lines the analytical ones. In (b) diamonds correspond to measured frequencies, and solid lines are the spline-interpolated calculated values.

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to the empty cavity, the TE_{011} mode exhibits the smallest frequency shift (about 2 GHz); the shift of the TE_{112} mode is twice as large. Additional increase in the dielectric load, for example, by filling the tube with the sample, results in a further separation of the two modes. Increasing of the tube radius alternates the order of the modes again; the downshift in frequency is larger for the TE_{011} mode than for the TE_{112} mode. With a tube with o.d. 3.8 mm, i.d. 2.9 mm, the resonance length for the TE_{011} mode is about 6.5 mm.

The observed trends are caused by the electric field distribution in the sample area, and the simulations give quite accurate numerical values for the trends. The eigenmode analysis shows that there exist ranges which are free from unwanted modes. This allows us to choose optimal operational conditions as listed in Table I. The ranges in the table are chosen to have the favored TE_{011} or TE_{012} mode in the middle of the range. Note that unwanted modes can be suppressed by cutting the pathways for the mw current along the z direction, for example, by scratching circular conducting layers, similar as described above for the ENDOR cavity, or adding mode suppression gaps on the ends of the CC. However, to some extent this affects also the TE_{01p} modes. On the one hand, a radiation leakage caused by the cutting slits reduces the merit factor. On the other hand, as studied in detail by Mett and Hyde,[22] the suppression gaps introduce a perturbation of the cavity eigenmodes leading to additional losses for the mode TE_{01p}. This is of particular importance for pulse EPR operation, conducted typically under overcoupling conditions and using broadband excitation.

The second step is to optimize the coupling structure for the TE_{011} mode. Intrinsically, the TE_{011} mode has closed **E** and **H** loops and the area where some components are at maximum cannot easily be used to provide the strongest coupling (see Fig. 2). Since overcoupling is an important issue in pulse EPR experiments, we studied this problem in more detail. A Gordon coupler[23] was chosen for stability reasons. To minimize incidental insertion losses the undercritical part of the waveguide is only 4 mm long. The coupler ends with a diaphragm, shown in Fig. 1, which has an iris with a width of 3.5 mm and a thickness of 0.12 mm. To make use of the maximum available access, the iris spans the range from the cavity wall to the sample access hole. For the modeling purpose the height of the iris was adjusted to reach critical coupling at the middle of the undercritical section, i.e., at about 2 mm away from the coupling window for the plunger in the Gordon coupler. The height was determined empirically by varying it in steps of 0.2 mm from 1 to 2 mm and found to be 1.2 mm for the 3 mm set with a 2.9 mm sample tube, silver plated CC, and a Gordon plunger with a dielectric constant of 3.4 (PEEK). This is our favorite set because it was found that a further increase in sample tube diameter does not increase the EPR signal, in agreement with Wang and Chasteen.[16] However, the possibility to utilize 3.8 mm tubes makes the probehead more versatile in use. Consequently, we optimized first the coupling for the 3 mm set and then developed a model which describes the observed values and trends with particular attention to the internal losses of the cavity. These simulations were carefully done and compared with the measured data. In a next step, the developed computer model was used to figure out how the coupling structure has to be modified for the 3.8 mm tubes (4 mm set) to get the same coupling range as for the 3 mm set. As expected, it was found that the coupling coefficient for constant cavity losses depends on (i) the dielectric constant of the plunger of the Gordon coupler, which affects the wave length in the waveguide, (ii) the entry hole diameter, which determines how close the end of the iris can be to the maximal radial component (about 2.5 mm off center for the empty cavity), and (iii) the sample tube radius, its wall thickness and dielectric constant, that modifies the "sucking effect" for the electric field.

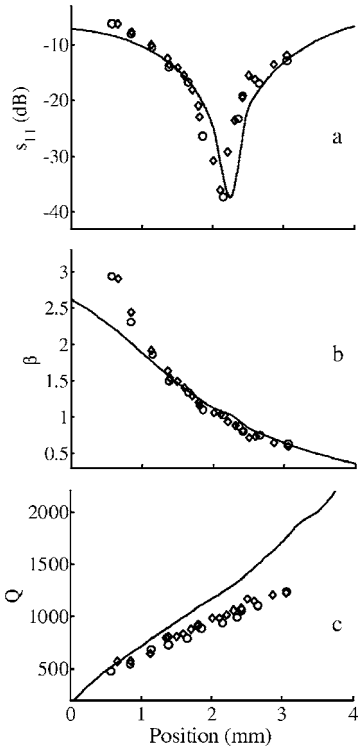


FIG. 4. Testing the structure. (a) Simulated (solid line) and measured (diamonds and circles) reflection coefficients as a function of the distance between the Gordon wedge and the coupling aperture. (b) Calculated coupling coefficients and (c) loaded merit factors.

Simulations and test measurements for the 3 mm set with the silver plated cavity are presented in Fig. 4. The simulations were performed using the model shown in Fig. 2, with the short replaced by a closed wall, a smaller entry hole (∅3.1 mm), and an iris height of 1.2 mm. The cavity response in the frequency range of 35 ± 0.5 GHz was calculated as a function of the Gordon plunger displacement, and the minimal s_{11} coefficients were picked up for each plunger position excluding the effect of the frequency shift. As shown in Fig. 4(a), the critical coupling is reached for a plunger displacement of 2.2 ± 0.2 mm. The mentioned deviation reflects an experimental error arising from the reassembly of the structure. Circles and diamonds denote successive measurements without the reassembly. About 100 000 mesh-cells have been used for the model calculation, and a conductivity of 6×10^7 S/m was assumed for the cavity walls. The quartz sample tube was treated as lossy material as well, with a tangent loss of 0.001 and a dielectric constant of 3.78.

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The tangent loss was the only parameter changed to "fit" the experimental data. It incorporates all the other possible losses (radiation leaks, moisture, etc.). The obtained value is approximately three times more than the one reported for the quartz at 35 GHz,[24] indicating that there are significant sources of loss which were not included in the simulation explicitly. In the following sections these sources will be discussed.

Calculations were done using adaptive meshing, thin sheet technology, and perfect boundary approximation options. The primary meshgrid in the iris plain had 0.3 mm resolution along the iris height. The convergence of the results has been checked using up to 200 000 meshcells, and shows that the resolution is accurate within the experimental uncertainty. The coupling coefficients were calculated from the s_{11} values using the equation $\beta=(1\pm|s_{11}|)/(1\mp|s_{11}|)$, where the upper sign refers to overcoupling and the lower sign to undercoupling. Simulated and measured coupling coefficients nearly coincide for $\beta<1.7$, as is shown in Fig. 4(b). For stronger couplings the experimental values exceed the simulated ones. This deviation is partially due to additional reflections caused by mechanical imperfections rising up when the Gordon plunger comes close to the iris, and the influence of S turn (Fig. 1) which was not included in the simulation.

The quality factors were calculated using the simple relation $Q_L=f/\Delta f$, where Δf is the width of the resonance curve at −3 dB level. The Q value for critical coupling was found to be about 950 (simulation gives 1150) [Fig. 4(c)]. For overcoupling the quality factor decreases to about 450. Experimental data points for distances shorter than 0.5 mm are skipped, because the observed s_{11} coefficient increased above −3 dB and it was not possible to use the same simple approach to calculate Q. The experimental data show a weaker dependence on the plunger position as compared to the simulation. This indicates again that additional losses exist in the real structure which were not included in the simulation. Candidates are a radiation leakage through sample holes, and the gap between the short and the sidewall. The latter is expected to be a suppressor for TE/TM_{mnp} modes with $m\neq 1$. The lack of symmetry in Fig. 2 suggests an existence of the current in the z direction which is a characteristic feature of these modes. Indeed, the close look on the corresponding current distribution shows that asymmetry is due to a small contamination of TM_{111} mode. Such contamination gives rise in the leakage and, consequently, decreases the Q value. A simple calculation (see as an example Ref. 25) shows that the Q values due to resistive and dielectric (using tangent loss=0.0003) losses are of the order of 15 000 each. To have the $Q\approx 1000$ at critical coupling the quality factor due to the leakage (and an imperfection of real structure) has to be about 3000. The later value is about two and half times less than total dielectric and resistive losses. However, the observed background signals at 4.2 K and at full power was found to be weak and no additional lossy material was introduced to the structure after the short.

The observations made with the 3 mm set demonstrate that our simplified model describes the real structure satisfactory. In a next step we modeled the 4 mm set with particular attention to the coupling coefficients. The model used for these computations is shown in Fig. 2. The field maximum in the coupling waveguide is shifted by a 0.45 mm more outwards field maximum into the cavity, as compared to the 3 mm set. This shift reduces the coupling, moving the PEEK plunger position for critical coupling towards the iris [case (a) in Fig. 5, 1.2 mm iris] by 1 mm. Increasing the aperture height to 1.6 mm improves the situation but the coupling is still smaller than for the 3 mm set [case (b) in Fig. 5]. A similar coupling is reached with an almost completely open waveguide (iris height=3 mm). However, a replacement of PEEK ($\varepsilon=3.4$) by teflon ($\varepsilon=2$) allows for the same coupling, even with a 1.2 mm aperture [case (c) in Fig. 5]. The latter effect can be explained as due to an elongation of the wavelength in the filled part of the waveguide (λ_g goes from 10 to 13 mm) and an interplay between the wave radiated on the plunger border and an evanescent wave. The same trend was observed experimentally; the teflon plunger showed a stronger coupling at the same position, however, the improvement was smaller than in the simulation. Conducted measurements and simulations demonstrate that the reachable overcoupling coefficient allows one to have the cavity bandwidth of about 80 MHz which is sufficient for the pulse EPR experiments used in applied spectroscopy.

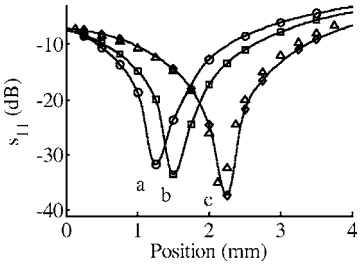


FIG. 5. Simulated dependence of the coupling for different Gordon plungers, aperture heights, and entry hole diameters. (a) PEEK plunger, 1.2 mm iris, ∅4 mm; (b) PEEK, 1.6 mm, ∅4 mm; (c) teflon, 1.2 mm, ∅4 mm (diamonds); PEEK, 1.2 mm, ∅3.1 mm (triangles).

A primary characterization of a cavity is its capability to generate the maximum magnetic field per unit incident power at the sample position. The simulations in Fig. 2 show that a mw power of $P=1$ W produces a magnetic field of $H=550$ A/m or a field induction in the rotating frame of $B_1=\mu_0 H/2\cong 0.35$ mT. The power-to-field conversion factor of the cavity normalized to the merit factor is $\Lambda\approx 0.01\ \mathrm{mT}/\sqrt{\mathrm{W}}$.

RESULTS

Representative examples of applications of the probehead can be found in the literature.[20,21] For instance, matched hyperfine sublevel correlation experiment (HYSCORE) was used to determine the spin density on the nitrogen ligand in $[Rh(trop_2N^{\bullet})(bipy)]^+$ trifluorosulfonate (Ref. 20), and the high efficiency of the probehead was used in hyperfine-decoupling experiments to unravel complicated electron spin echo envelope modulation (ESEEM) spectra.[21] Here, we concentrate on experimental results to characterize the performance of the probehead, in particular, the mw and rf conversion factors and the sensitivity.

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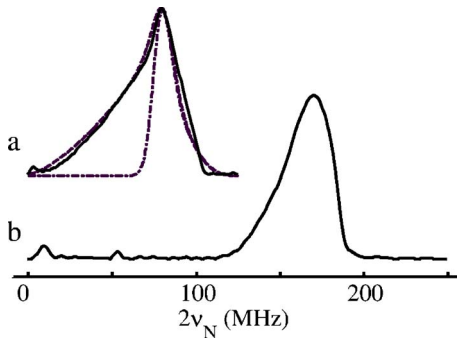


FIG. 6. Experimental determination of the magnetic field induction in the cavity. Fourier transforms of the nutation signals. (a) γ-irradiated quartz powder in a 2.9 mm sample tube. (b) 1-mm-thick single crystal of $Zn(Im)_6$:1% Mn^{2+} positioned close to the middle of resonating volume.

Test measurements were performed on a bench setup with the 3 mm set. To determine the mw field strength, the transient behavior of the electron spins during a mw pulse was recorded. The magnetic field $\mathbf{B}_1$, oscillating at the precession frequency, forces the resonant spins to nutate around $\mathbf{B}_1$ at angular frequency $\omega_N = (g\mu_\beta B_1/\hbar)\sqrt{S(S+1)-m_s(m_s+1)}$, where S and m_s are the electron spin and magnetic quantum numbers. If mw and the resonance frequency differ by the resonance offset Ω_s, the effective nutation frequency is given by $\omega_N^{\text{eff}} = \sqrt{\omega_N^2+\Omega_s^2}$. For inhomogeneous lines the integrated signal is a function of Ω_s and the B_1-field distribution.

For the measurement of the mw field amplitude, the phase-inverted echo-amplitude detected (PEANUT) sequence[26] was applied to γ-irradiated quartz (E' centers). The sample tube with a diameter of 2.9 mm was filled with fine-crushed quartz powder. The pulse sequence was $t_{(\pi/2)x} - \tau - t_{(\text{HTA})x} - T - t_{(\text{HTA})-x} - \tau -$ echo, where $t_{(\pi/2)x}$ is the length of a pulse with flip angle $\pi/2$ along the x axis of the rotating frame; $t_{(\text{HTA})x}$ is the length of a high-turning angle (HTA) pulse, and $T < (t_{(\text{HTA})x} + t_{(\text{HTA})-x})$ is the variable time of phase inversion. After Fourier transformation the nutation spectrum is obtained. The measured nutation frequencies are doubled because the sequence is based on a rotary echo, which refocuses partially the B_1-field inhomogeneity.

At full mw power for the HTA pulse (42 W at the probehead input), the maximum nutation frequency $2\omega_N/2\pi$ is about 80 MHz for the γ-irradiated quartz in a slightly overcoupled cavity, as is shown in Fig. 6(a). The low-frequency tail in the spectrum is due to the $\sin(2k_z z)$ variation in the mw field along the z axis of the cavity (H_z component in Fig. 2). This variation was used to simulate the nutation spectrum [dashed line in Fig. 6(a)]. The simulation was based on the model used by Stoll *et al.*[26] and includes offset and linewidth effects as well. For comparison, a constant-field simulation is also shown in Fig. 6(a) (dash-dotted line). Corresponding measurements with a small single crystal of $Zn(Im)_6$:0.4% Mn^{2+} ($S=5/2$), placed near the cavity center, show much sharper, relatively, nutation line for the $m_s = 3/2 \rightarrow 5/2$ transition, with an ω_N value which is a factor $\sqrt{5}$ larger than for $S=1/2$ systems theoretically. From the nutation measurements we found a conversion factor $\Lambda = (9.3 \pm 0.5) \times 10^{-3}$ mT/$\sqrt{\text{W}}$. The deviation between the measured conversion factor and the value obtained by the simulation is smaller than 10%. For comparison conversion factors reported for commercial probeheads are given in Table II, showing that the efficiency of our setup is in between the sapphire ring resonator for 2 mm samples and the cylindrical 3 mm setup for cw EPR.

TABLE II. Probehead comparison.

Model	ER5107D2[a]	This work	ER5106QT[a]
Tube o.d. (mm)	2	3.8	3
Λ, 10^{-2} mT/$\sqrt{\text{W}}$	2	0.9	0.6
Bandwidth (MHz)	36	40	13

[a]Reference 4.

Nutations of protons under a rf field have been used to estimate the conversion factor of the rf coil. A Mims ENDOR experiment[27] was carried out on coal as a function of the rf pulse length. The rf coil terminated by a 50 Ω load was matched to the source impedance at the proton resonance frequency (about 50 MHz), using a 33 pF capacitor in series. The circuit bandwidth was about 50 MHz, providing $Q \cong 1$. The observed Rabi oscillation induced by a rf pulse of 300 W is shown in Fig. 7. The first minimum of the oscillation is observed at 26 μs and corresponds to a nutation angle of 180°, resulting in a conversion factor of $\Lambda_{\text{rf}} = B_2/\sqrt{PQ} \approx 1.5 \times 10^{-3}$ mT/$\sqrt{\text{W}}$. This relatively low value is expected because the saddle coils are separated by about 13 mm and the rf field in the center of the coils is roughly inversely proportional to their separation.

The low conversion factor requires relatively high rf power. As a consequence, heating effects are observed when a helium flow cryostat is used. They manifest in slight changes of the resonance conditions and in phase shifts. For example, the bump at about 35 MHz in the proton ENDOR spectrum of a glycine single crystal doped with 1% Cu^{2+}, shown in the inset in Fig. 7(a), is due to such a heating effect. Note that the heating problem is quite common in pulse ENDOR; it can be removed by simply varying the rf at random.[28]

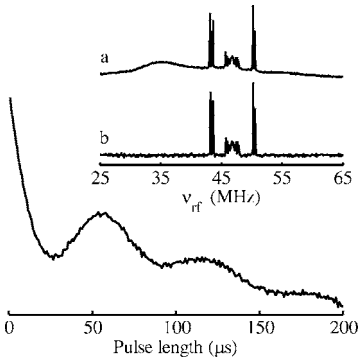


FIG. 7. Rabi oscillations observed in proton ENDOR of coal. The position of the first minimum (pulse length 26 μs) corresponds to a 180° rotation angle for the proton spins. A low-Q, matched rf coil terminated to the ground via 50 Ω has been used. Insets show proton ENDOR spectra of Cu^{2+} ions doped into an α-glycine single crystal. The spectra were measured at 15 K using 450 W of rf power; pulses of 25 μs were applied to an unmatched rf coil. (a) A linear sweep of the rf; introduces base line distortions due to heating effect. (b) rf applied randomly providing a flat base line.

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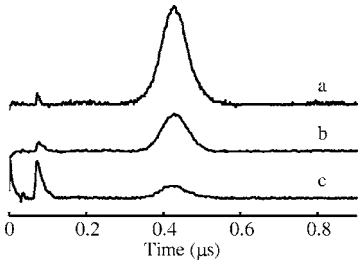


FIG. 8. Single shot electron spin echoes of γ-irradiated quartz powder in sample tubes of different diameters. (a) i.d. 2 mm, relative mw amplitude is −15 dB, (b) i.d. 1 mm, −17 dB, and (c) i.d. 1 mm with ER5106-QTE probehead (Bruker), −7.5 dB. Vertical scale setting of the digital oscilloscope (LeCroy 520) was 0.5 V/div (a), and 0.2 V/div [(b) and (c)]. The amplitude ratios of the observed signals are $a:b:c=8:3:1$.

Finally, a sensitivity test has been carried out in the pulse mode. The spin echo signal of a small amount of γ-irradiated quartz (3.5 mg, 1 cm long in an i.d. 1 mm tube), induced by the sequence 16 ns-τ-32 ns, was measured. The observed signal-to-noise ratio was about eight for a receiver bandwidth of 200 MHz. The radical concentration, 2.2×10^{17} spins/cm^3, was calculated using the concentration-dependent instantaneous diffusion of the spin echo measured at *X* band following the procedure described in Ref. 29. From these data the minimum detectable number of spins was calculated, yielding 3×10^9 spins/$\sqrt{\text{Hz}}$, which serves as an upper limit for that quantity, because the effect of partial line excitation was not considered. This value nearly coincides with the sensitivity reported for the Varian CC[15]. To provide a feeling for the signal strength we measured echoes on E' centers using different sample tubes: i.d. 1 mm for the Bruker ENDOR probe head (ER5106QTE) and i.d. 1 and 2 mm for our setup. In all these cases, the conditions were optimized for maximal signals. Figure 8 demonstrates that for the same tube the signal is three times higher and with the larger tube the echo is eight times stronger than the one obtained with the ER5106QTE probehead. The required mw power was −(17/15) dB for our probehead and −8 dB for the Bruker one. The signals were measured on a digital oscilloscope operating in the single shot mode. A larger scale (0.5 V/division) has been used for the strongest signal, therefore a digitizing noise becomes visible in case (a). The spikes at the beginning of the time axis are ringing of the cavities. Zero time coincides with the end of the last mw pulse.

Experiments with the 4 mm set were performed as well, showing a slight degradation in the cavity conversion factor and the similar sensitivity. The probehead operates stably at the temperatures down to 4.2 K. At the helium temperature the quality factor increased by a factor of two, but the reachable coupling coefficient (about 3) allows one to have about the same cavity ringing time as was obtained at room temperature (see Fig. 8). The relatively high quality factor in connection with the rather small achievable coupling expected for cylindrical cavities is found to be not a serious limitation for pulse experiments. It is worth mentioning that the coupling can easily be increased using, for example, a loop coupling. However, this diminishes two major benefits of the described setup: a simplicity in construction and robustness in use. Together with the possibility to use large sample tubes, the developed probehead provides a nearly optimal configuration for a wide variety of routine and advanced experiments at Ka band.

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